

HEMANTH VASUDEV N P

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EDUCATION

PSG College of Technology
M.Sc (Integrated) Software Systems

Present
CGPA: 8.43 / 10

The TVS School, Madurai
High School

June 2024
Percentage: 90.3

SKILLS

Languages: Python, C++, C, JavaScript
Frameworks: FastAPI, Flask, React.js, Node.js, Express.js
Libraries: scikit-learn, NumPy, SciPy, sentence-transformers, asyncio
Databases: PostgreSQL, MongoDB, MySQL, Redis
DevOps: Docker, Linux, Git, Postman

PROJECTS

CineScope — Film Discovery & Recommendation Platform |  Code |  Demo

- **Groq LLaMA 3.1** function calling classifies natural-language queries into 8 intent types and 10 entity fields — replaces brittle keyword matching with structured JSON extraction.
- Hybrid recommender: **TF-IDF** cosine similarity + **TruncatedSVD** collaborative filtering with tiered fallback (hybrid → content → genre → trending) based on available user signal.
- Parallel service calls via **asyncio** with per-service **circuit breaker** fallback chain; 71 pytest tests gate a GitHub Actions CI/CD pipeline across 5 Dockerized services.

Throttlr — High-Performance API Gateway |  Code |  Demo

- **C++20** API gateway on **Linux** with **epoll** event loop and worker thread pool for concurrent request handling.
- Solved distributed rate-limit races with an atomic **Redis Lua** script (vs. naive GET→SET); wait-free **circuit breaker** via **compare_exchange_strong** for zero-mutex state transitions on the hot path.
- **SIGHUP** config reload with zero dropped connections; four load balancing strategies including consistent hash (150 virtual nodes).

MargaMetis — Intelligent Route Optimizer |  Code |  Demo

- Graph engine on a 22K-node Chennai **OSM** graph — **Dijkstra**, **A***, **Bidirectional A***, and **Yen's K-Shortest** from scratch; Bidirectional A* explored **12.9× fewer nodes** than Dijkstra (6ms vs 60ms).
- Cost-function injection architecture: each routing mode supplies a **(u,v,data)→float** callable injected at traversal time — decouples routing logic from cost policy, **O(1)** mode switching with zero graph mutation.
- **Groq LLaMA 3.1** few-shot prompting parses free-text queries into structured JSON cost weights; **Redis** caching cut repeat-query latency from **3,500ms to 16ms**.

ACTIVITIES & HOBBIES

CSA Tech Team Tech Team member of the Computational Sciences Association (CSA), PSGCT

FinVerse Analyst on PSGCT's FinVerse finance club

Photography Amateur photography with a focus on composition and detail